Eric Liu

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EDUCATION:

University of Texas at Austin - Mechanical Engineer, Expected Graduation Date: May 2028

SKILLS:

Computer Skills: Ubuntu Linux, Arduino, Raspberry Pi, ESP32, Javascript, HTML, Python, Bash, MATLAB
Technical Skills: Soldering, manual machining, 3D printing, PCB design (KiCad), CAD in FreeCAD, Onshape, and SolidWorks, CFD, FEA, and thermal simulations in ANSYS
Languages: Fluent in Chinese and English

EXPERIENCE:

Texas Design Build Fly (September 2024 - May 2026) | *Systems Lead Engineer (August 2025 - May 2026)*

- Directed a year-long project lifecycle, leading a 15 member systems team to design and manufacture payload mechanisms, conduct structural and reliability testing, and research process and material improvements
- Streamlined manufacturing schedule and team communication through task decomposition and translating high-level leadership decisions into actionable engineering objectives, increasing team retention by 60%
- Iterated 20+ versions of lightweight UAV circuitry to minimize mass, designed hardware testing and validation routine including mechanical drop tests and oscilloscope based electrical tests to ensure reliability
- Developed custom test setups and sensor instrumentation, including installing load cells and pitot tubes, to characterize towed banner aerodynamics, providing critical stability and drag data
  - Analyzed data collected from across 20+ vehicle based tests, wind tunnel tests, and flight tests in Python
  - Performed root cause analysis of banner deployment failures, iterating folding geometry, banner rigging, and release mechanisms to improve deployment reliability

IEEE Robotics and Automation Society (September 2024 - May 2025)

- Led a team to design and assemble the gantry for a chess playing robot utilizing a CoreXY design, reducing size of the robot by 50% compared to its predecessor while increasing movement speed
- Managed documentation, progress, and deadlines, allowing the team to complete the gantry on time

Research with University of Shanghai for Science and Technology (January 2026 - May 2026)

- Performed CFD modelling of gas flow inside an atomic layer deposition reaction chamber with ANSYS Fluent
- Optimized parameters such as injection times and chamber geometry to minimize reactant use and improve manufacturing speed, validated results through cross-functional collaboration with experimental team

PROJECTS:

Remote Controlled Vehicles:

- Remote controlled (RC) aircraft (June 2022 - Present)
  - Built a total of 9 aircraft, each with different goals such as being able to perform short take off and landing (STOL), being easy to control, or being able to carry a large payload
  - Conducted ANSYS Fluent CFD simulations of low Reynolds number airfoils and modeled fluid structure interaction with FEA to optimize wing size, then verified simulation results with static structural testing
- RC Boat with a crane attached to a camera module and neodymium magnet to take underwater pictures and retrieve underwater metal objects at a depth of 50 ft (May - December 2023)
  - Used an ESP8266 microcontroller to control motor throttle and raising/lowering the camera module

Coding, Electronics, and AI:

- Built a Chrome extension that added dark mode and grade tracking to the Skyward gradebook system, used by 100+ students (April, 2021 - August, 2023)
- USB charge limiter for an Android phone using an ESP32 microcontroller to connect to the phone via bluetooth, triggering a relay to cut power when a certain battery percentage is met (July 2025)
- Built an AI writing detector that achieved 95% accuracy in AI detection benchmarks (February - August, 2025)

Eye Tracking Project: (September 2025 - Present)

- Designed and prototyped glasses frames with integrated IR cameras and designed video storage electronics
- Participated in cross-functional design meetings to ensure integration with software and electronics teams
- Utilized design for manufacturing principles to optimize glasses frames for 3D printing